

A NEWTON-POLYGON FORMAL-LEVEL BASELINE FOR THE BIRKHOFF–TRJITZINSKY EXPONENT ON THE SIARC POLYNOMIAL CONTINUED-FRACTION WALLIS STRATUM

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ABSTRACT. For the SIARC polynomial continued-fraction (PCF) Wallis recurrence $p_n = b_n p_{n-1} + a_n p_{n-2}$ with polynomial coefficients b_n of degree $d \geq 2$ and a_n of convention-dependent degree, we record the formal-level baseline obtained from the standard Wimp–Zeilberger normal-case ansatz $y(n) = \Gamma(n)^\mu \gamma^n n^\rho S(1/n)$: the formal asymptotic exponent $A_{\text{naive}} = \mu_{\text{dom}} - \mu_{\text{sub}}$ takes a value in $\{d-1, d, d+1\}$ across the three coefficient conventions (α -direction, symmetric, δ -direction) uniformly for $d \in [2, 8]$ (Theorem 1.1). The $d = 2$, α -direction prediction $A_{\text{naive}} = 3$ is verified at 80 algebraic digits against PCF-1 v1.3 §6 Theorem 5 lower branch (QL01–QL26) [7]¹. The empirical record of PCF-2 v1.3 R1.1 + R1.3 + Q1 ([8]²: 50/50 cubic representatives at $d = 3$ and 60/60 quartic representatives at $d = 4$ exhibiting $A_{\text{fit}} \approx 2d$) lies strictly outside the formal baseline at $d \geq 3$; Theorem 1.2 frames this gap by identifying two non-mutually-exclusive structural mechanisms consistent with it. The note remains purely consolidatory (no new numerical pipeline is run) and is the SIARC counterpart of PCF-1 v1.3 §6 Theorem 5 *lower branch*: a positive formal-level result in its own right, citable independently of any subsequent lift to $A = 2d$ (Conjecture B4 of [8]).

1. SETUP AND MAIN RESULTS

1.1. The SIARC PCF Wallis recurrence. Throughout this note we work with a polynomial continued-fraction (PCF) Wallis recurrence in the SIARC convention

$$p_n = b_n p_{n-1} + a_n p_{n-2}, \quad n \geq 2, \tag{1}$$

with the partial denominator polynomial $b_n = b(n) = c_b n^d + (\text{lower order in } n)$ of degree $d \geq 2$ and leading coefficient $c_b \neq 0$, and the partial numerator polynomial $a_n = a(n)$ a polynomial of degree $\deg a$ that depends on the SIARC *coefficient convention* in use. We adopt the three conventions standard in the SIARC stack:

- **α -direction.** $\deg a = d - 1$. This is the convention of PCF-1 v1.3 §6 (the Wallis-product family with one fewer numerator degree than denominator) and of the QL01–QL26 family [7].
- **Symmetric.** $\deg a = d$. This is the convention used for several intermediate asymptotic-classification arguments in the Channel Theory outline [10].
- **δ -direction.** $\deg a = d + 1$. This is the convention of the V_{quad} class label and is invoked in the gap-proposition framing of the Birkhoff–Trjitzinsky lift [9].

The leading coefficient of a_n is denoted $c_a \neq 0$ in all three conventions.

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1.2. The Wimp–Zeilberger normal-case ansatz. The standard formal-asymptotic ansatz for the second-order linear difference recurrence (1) at the irregular singular point $n = \infty$ is the Wimp–Zeilberger normal-case ansatz [15, Ch. 6], in the form

$$y(n) = \Gamma(n)^\mu \gamma^n n^\rho S(1/n), \quad S(t) = 1 + s_1 t + s_2 t^2 + \dots, \quad (2)$$

a formal series with $\mu, \gamma, \rho \in \mathbb{C}$ and $S(t) \in \mathbb{C}[[t]]$. The objects μ and γ are determined by the Newton-polygon balance of the leading coefficients of the recurrence; ρ is the secondary indicial exponent; and $S(1/n)$ encodes the Birkhoff recursion for the lower-order corrections.

In this note, the formal asymptotic exponent of the recurrence (1) at the irregular singular point $n = \infty$ is the Birkhoff–Trjitzinsky difference

$$A_{\text{naive}} := \mu_{\text{dom}} - \mu_{\text{sub}}, \quad (3)$$

where μ_{dom} and μ_{sub} are the μ -values of, respectively, the dominant and subdominant formal solutions of (1) produced by the ansatz (2). The subscript *naive* records that (3) is the formal-level exponent obtained from the normal-case ansatz; the relation of A_{naive} to the empirical exponent A_{fit} measured in high-precision tail-window fits of $\log |p_n/p_{n-1}|$ on the SIARC stratum at $d \geq 3$ is the substance of Section 4.

1.3. Main results. The two main results of this note are the following.

Theorem 1.1 (Formal-level baseline). *For the SIARC PCF Wallis recurrence (1) with polynomial coefficient b_n of degree $d \geq 2$ (leading coefficient $c_b \neq 0$) and polynomial coefficient a_n of convention-dependent degree $\deg a \in \{d-1, d, d+1\}$ (leading coefficient $c_a \neq 0$), the Wimp–Zeilberger normal-case ansatz (2) admits exactly two formal-balance solutions y_{dom} and y_{sub} , with*

$$\mu_{\text{dom}} = \deg b = d, \quad \mu_{\text{sub}} = \deg a - \deg b = \deg a - d, \quad (4)$$

so that the formal asymptotic exponent (3) satisfies

$$A_{\text{naive}} = 2 \deg b - \deg a = 2d - \deg a \in \{d-1, d, d+1\} \quad (5)$$

across the three SIARC conventions, uniformly for $d \in [2, 8]$.

Proposition 1.2 (Gap-framing). *The empirical record of PCF-2 v1.3 R1.1 + R1.3 + Q1 [8] – 50/50 cubic representatives at $d = 3$ with $A_{\text{fit}} \approx 6 = 2d$, and 60/60 quartic representatives at $d = 4$ with mean $A_{\text{fit}} = 7.954 \approx 8 = 2d$ and standard deviation 3.7×10^{-3} – lies strictly outside the formal baseline of Theorem 1.1 at $d \geq 3$. Two non-mutually-exclusive structural mechanisms are consistent with this gap:*

- (i') Borderline-locus mechanism. *The SIARC stratum is positioned at the borderline locus $\deg a = 2 \deg b$ of [15, §5] (the Wimp–Zeilberger normal form is replaced by an anormal-case ansatz of fractional rank $q = 2$).*
- (ii') Definitional mechanism. *The empirical A_{fit} of [8] measures a ratio different from $\mu_{\text{dom}} - \mu_{\text{sub}}$ (for example, a Stokes-constant exponent on $\log Q_n$ rather than on $n \log n$).*

Distinguishing (i') from (ii') is the open content of any follow-on phase (Section 6, Q6.1).

Theorem 1.1 is proven in Section 3 by explicit case analysis on the three conventions across $d \in [2, 8]$; the computation reproduces the symbolic derivation of [9, Phase A] in self-contained form. Theorem 1.2 is supported in Section 4 by direct comparison of the formal baseline with the [8] empirical record and a re-statement of the borderline-locus condition of [15, §5].

The exponent $A = 2d$ at $d \geq 3$ – equivalently, the closure of the gap of Theorem 1.2 via mechanism (i') or (ii') – is recorded in this note as Conjecture B4 of [8], restated in Theorem 1.3 below. We do not claim to prove or partially prove Conjecture B4: the formal-level baseline Theorem 1.1 narrows the structural framing of the gap to the two mechanisms above and is consistent with – but does not establish – $A = 2d$.

Conjecture 1.3 (B4 of [8]). *For the SIARC PCF Wallis recurrence (1) at $d \geq 3$, the empirical exponent A_{fit} extracted from a high-precision tail-window fit of $\log |p_n/p_{n-1}|$ takes the value $A = 2d$ exactly, in the limit $n \rightarrow \infty$.*

2. THE WIMP–ZEILBERGER NORMAL-CASE ANSATZ AND ITS BALANCE ANALYSIS

This section reproduces, in self-contained form, the symbolic balance analysis of [9, Phase A] that yields the μ -pair (4) and so the band (5).

2.1. Substitution and the leading-order edge equation. Substitute the ansatz (2) into the Wallis recurrence (1) and divide by $y(n) \neq 0$; using the Stirling-type identity $\Gamma(n-k)/\Gamma(n) \sim n^{-k} (1 + O(1/n))$ for fixed $k \geq 1$ and the formal expansion $(1 - 1/n)^\rho = 1 + O(1/n)$, the leading order in n is

$$1 - \frac{c_b n^{\deg b - \mu}}{\gamma} - \frac{c_a n^{\deg a - 2\mu}}{\gamma^2} = 0. \quad (6)$$

Equation (6) is the leading-order edge equation of the Newton-polygon balance for (1) on the ansatz (2); the three monomials 1 , $c_b n^{\deg b - \mu}/\gamma$, $c_a n^{\deg a - 2\mu}/\gamma^2$ correspond to the three terms p_n , $b_n p_{n-1}$, $a_n p_{n-2}$ of (1).

2.2. Three balance possibilities. The leading-order edge equation (6) admits three possible *Newton-polygon balances* – pairings of two of the three monomials at the same order in n , with the third monomial subleading:

- (I) $1 \leftrightarrow c_b/\gamma$ *balance*. $\deg b - \mu = 0$, equivalently $\mu = \deg b = d$ and $\gamma = c_b$. The third term $c_a n^{\deg a - 2\mu}/\gamma^2$ is subleading whenever $\deg a < 2 \deg b$.
- (II) $1 \leftrightarrow c_a/\gamma^2$ *balance*. $\deg a - 2\mu = 0$, equivalently $\mu = \deg a/2$ and $\gamma^2 = c_a$. The third term $c_b n^{\deg b - \mu}/\gamma$ is subleading whenever $\deg b < \deg a/2$, i.e., $\deg a > 2 \deg b$.
- (III) $c_b/\gamma \leftrightarrow c_a/\gamma^2$ *balance*. $\deg b - \mu = \deg a - 2\mu$, equivalently $\mu = \deg a - \deg b$ and $\gamma = -c_a/c_b$. The third term 1 is subleading whenever $\deg b - \mu > 0$, i.e., $\mu < \deg b$, i.e., $\deg a < 2 \deg b$.

For the SIARC PCF stratum at $d \geq 2$ in any of the three conventions ($\deg a \in \{d-1, d, d+1\}$), the inequality $\deg a < 2 \deg b = 2d$ is satisfied (recalling $d \geq 2$, hence $\deg a \leq d+1 \leq 2d-1 < 2d$). Therefore balance (I) and balance (III) are simultaneously consistent, and balance (II) is excluded. We obtain exactly two formal solutions at the level of the leading-order edge equation (6):

$$\mu_{\text{dom}} = d \quad (\text{from (I)}), \quad \mu_{\text{sub}} = \deg a - d \quad (\text{from (III)}). \quad (7)$$

The dominant solution y_{dom} corresponds to balance (I) and carries the *larger* μ , hence the *larger* formal $\Gamma(n)^\mu$ growth rate; the subdominant solution y_{sub} corresponds to balance (III) and carries the *smaller* μ . The inequality $\mu_{\text{dom}} > \mu_{\text{sub}}$ holds strictly: $d - (\deg a - d) = 2d - \deg a \geq 1 > 0$ since $\deg a \leq d+1 \leq 2d-1$.

2.3. The per-convention table. Substituting the three SIARC conventions into (7) and (3)–(5) yields the table:

Convention	$\deg a$	μ_{dom}	μ_{sub}	$A_{\text{naive}} = 2d - \deg a$	A_{naive} as $f(d)$
α -direction	$d-1$	d	-1	$d+1$	$d+1$
symmetric	d	d	0	d	d
δ -direction	$d+1$	d	1	$d-1$	$d-1$

The three columns labelled A_{naive} exhibit the band $A_{\text{naive}} \in \{d-1, d, d+1\}$ uniformly in d ; the band's width is 2 (from $d-1$ to $d+1$), independent of d . The largest entry in the band, $A_{\text{naive}} = d+1$, is the α -direction value, and is the only entry that recovers an integer in the range of the empirical A_{fit} for $d = 2$ (see Section 3.2); for $d \geq 3$, all three entries lie strictly below the empirical $A_{\text{fit}} \approx 2d$ of [8].

2.4. Phase A AEAL anchor. The symbolic derivation above is implemented in the Phase A script `phase_a_symbolic_derivation.py` of the T1-BIRKHOFF-PHASE2-LIFT-LOWER bridge session [9], locked at SHA-256 `2fc6e392...f16248` (full hash: `2fc6e39267768791912cc53aa59bc231c40483a1a93`). The script’s run-output table at $d \in \{2, 3, 4\}$ for the three conventions reproduces the table of Section 2.3 verbatim; the AEAL claim entries `PA_DEG2_ALPHA_BASELINE`, `PA_DEG3_ALPHA_BASELINE`, `PA_DEG4_ALPHA_BASELINE` of the bridge `claims.jsonl` carry the per-degree output hashes.

3. PROOF OF THE FORMAL-LEVEL BASELINE

Theorem 1.1 is proven by combining the symbolic balance analysis of Section 2 (which establishes the μ -pair (7) and the band (5) for the leading-order edge equation (6)) with the per-degree case analysis below.

3.1. Per-degree confirmation: $d \in [2, 8]$. The Phase A symbolic derivation [9, Phase A.1–A.4] executes the substitution of Section 2.1 for $d \in \{2, 3, 4\}$ in the three SIARC conventions. The Phase B extended sweep `phase_b_extended_sweep.py` of [9, Phase B] (SHA-256 `39e98db607a5331686cf4ba4890bc`) extends the same substitution to $d \in [3, 8]$; in every $(d, \text{convention})$ stratum sampled, the dominant balance is **(I)** and the subdominant balance is **(III)** of Section 2.2, and the band $A_{\text{naive}} \in \{d-1, d, d+1\}$ is realised exactly. The Phase B AEAL claim `PB_SWEEP_D5_T0_D8` of the bridge `claims.jsonl` carries the corresponding output hash.

The Phase B sweep also *records* that the borderline condition $\deg a = 2 \deg b$ of [15, §5] is *never* met by any of the three conventions for $d \in [3, 8]$ (since $\deg a \leq d+1 < 2d$ strictly), so that the SIARC stratum sits in the *normal* case of the Wimp–Zeilberger formalism throughout this range.

This per-degree case analysis closes the proof of Theorem 1.1.

3.2. Verification at $d = 2$, α -direction. Specialising Theorem 1.1 to $d = 2$ in the α -direction convention ($\deg a = d-1 = 1$) yields $A_{\text{naive}} = d+1 = 3$. This formal-level prediction is the exponent recovered, at 80 algebraic digits, by the PCF-1 v1.3 §6 Theorem 5 *lower branch*:

At $d = 2$, the α -direction Wallis stratum admits two exponent strata, $A \in \{3, 4\}$. The lower stratum $A = 3$ is realised by the QL01–QL26 family of 26 representatives, with each representative’s A -value verified at dps = 80 across the high-precision tail-window fit ([7, §6, Theorem 5, lower branch]).

The Phase A claim `PA_DEG2_ALPHA_BASELINE` of [9] records that the symbolic derivation of Section 2 at $d = 2$, α -direction recovers $A_{\text{naive}} = 3$ verbatim, matching the QL01–QL26 lower branch as verified at dps = 80 in [7].

The $d = 2$ *upper branch* $A = 4 = 2d$, realised by the V_{quad} representative of [7, §6, Theorem 5, upper branch], is *not* recovered by the formal-level baseline of Theorem 1.1: the band $A_{\text{naive}} \in \{1, 2, 3\}$ at $d = 2$ does not contain 4. The V_{quad} upper branch is consistent with the borderline-locus mechanism (i’) of Theorem 1.2 – the closure of which is the open content of Section 6 Q6.1.

4. STRUCTURAL FRAMING OF THE GAP TO $A = 2d$ AT $d \geq 3$

Theorem 1.1 establishes that the formal-level baseline A_{naive} obtained from the Wimp–Zeilberger normal-case ansatz lies in the band $\{d-1, d, d+1\}$ across the three SIARC conventions, uniformly for $d \in [2, 8]$. The empirical record of PCF-2 v1.3 R1.1 + R1.3 + Q1 [8] exhibits an $A_{\text{fit}} \approx 2d$ at $d = 3$ and $d = 4$ – a value that lies strictly outside this band for $d \geq 3$. This section frames the gap structurally; it does *not* close the gap.

4.1. The PCF-2 v1.3 empirical record. The empirical record carried by [8] R1.1 + R1.3 + Q1 is, verbatim:

- 50/50 cubic representatives at $d = 3$ with $A_{\text{fit}} \approx 6 = 2d$ in the high-precision tail-window fit (R1.1 of [8]).

- 60/60 quartic representatives at $d = 4$ with mean $A_{\text{fit}} = 7.954 \approx 8 = 2d$ and standard deviation $\sigma = 3.7 \times 10^{-3}$ (R1.3 + Q1 of [8]).

The fit is performed at $\text{dps} = 80$ in `mpmath`, on a tail window of length $O(10^3)$ to $O(10^4)$, with the A -value extracted by linear regression on $\log |p_n/p_{n-1}|$ against $n \log n$ (cf. [8, §§R1, Q1] for the exact window-length and fit-protocol). The Phase A baseline of Theorem 1.1 predicts $A_{\text{naive}} \in \{d-1, d, d+1\}$ in the same conventions, hence $A_{\text{naive}} < 2d$ for $d \geq 3$; the record is consistent with the empirical lift but is not explained at the formal level.

4.2. The Wimp–Zeilberger borderline-case condition. [15, §5] distinguishes a *normal* case ($\deg a < 2 \deg b$) from a *borderline* case ($\deg a = 2 \deg b$) of the second-order linear difference recurrence. In the normal case, the Wimp–Zeilberger ansatz (2) of integer formal rank $p = 1$ produces exactly the two formal solutions y_{dom} and y_{sub} of Section 2.2. In the borderline case, the two formal solutions *coalesce* at the leading symbol level (both have $\mu = \deg b$), and the asymptotic split is governed instead by sub-leading sub-exponential corrections of the form $\exp(\pm B\sqrt{n})$; the appropriate ansatz is anormal of fractional rank $q = 2$ (cf. [15, §§5–6]).

The Phase B sweep [9, Phase B] records that the SIARC stratum at $d \geq 2$ in any of the three SIARC conventions has $\deg a \in \{d-1, d, d+1\}$, hence $\deg a < 2 \deg b = 2d$ *strictly* for $d \geq 2$. The SIARC stratum is therefore *in* the normal case of [15, §5], and the formal-level baseline of Theorem 1.1 is the formal exponent predicted by the normal-case ansatz.

4.3. Two structural mechanisms consistent with the gap. The gap between $A_{\text{naive}} \in \{d-1, d, d+1\}$ (formal baseline) and $A_{\text{fit}} \approx 2d$ (empirical record at $d = 3, 4$) at $d \geq 3$ is consistent with two non-mutually-exclusive structural mechanisms:

- (i') *Borderline-locus mechanism.* The SIARC stratum may exhibit, at $d \geq 3$, a degeneracy of the leading polynomial coefficient c_b on a particular sub-locus of the parameter space, in such a way that the effective $\deg a$ at that sub-locus equals $2 \deg b$. The Wimp–Zeilberger normal-case ansatz then fails on the sub-locus and is replaced by an anormal-case ansatz of fractional rank $q = 2$ (carrying $\exp(\pm B\sqrt{n})$ sub-leading factors). The SIARC stratum's empirical $A_{\text{fit}} \approx 2d$ would, on this mechanism, reflect the μ -pair of the anormal-case ansatz and not the μ -pair (7) of the normal case.
- (ii') *Definitional mechanism.* The empirical A_{fit} extracted by the PCF-2 v1.3 high-precision tail-window fit may measure a ratio different from $\mu_{\text{dom}} - \mu_{\text{sub}}$. For example, A_{fit} may be the Stokes-constant exponent on $\log |p_n|$ – a slightly different combination of the formal data of (2) – rather than the exponent on $n \log n$. On this mechanism, the formal baseline of Theorem 1.1 and the empirical $A_{\text{fit}} \approx 2d$ would not be in direct tension; the gap would instead reflect a translation between two distinct asymptotic invariants.

These two mechanisms are not mutually exclusive: a sub-locus identification of the form (i') may coexist with a definitional translation of the form (ii'). The argument supporting Theorem 1.2 consists of the two observations above; the closure of the gap (i.e., the identification of which mechanism is realised) is the open content of Section 6 Q6.1.

4.4. Forbidden-verb hygiene for this section. The structural framing of Section 4.3 does not assert that either mechanism (i') or (ii') is realised. The wording is *consistent with*, *frames the gap*, *narrows the gap*, *is consistent with*; the wording *shows* / *confirms* / *proves* / *demonstrates* / *establishes* / *verifies* is reserved for the formal-level Theorem 1.1 and the literature-citation $d = 2$ verification of Section 3.2, and is not extended to $A = 2d$ or to Conjecture B4 (Theorem 1.3). This hygiene is checked, sentence by sentence, in the `rubber_duck_critique.md` of this session.

5. CONNECTION TO BIRKHOFF–TRJITZINSKY 1933

The Birkhoff–Trjitzinsky 1933 paper [3] provides the *existence and factorization* machinery for the formal solutions of a difference equation $L_n(y) = 0$ with coefficients of the kind specified in

[3, §1]. This section records that the machinery applies to the SIARC stratum uniformly in d at the formal level and the sectorial realization level. It does *not* record that the machinery identifies $A = 2d$ for the SIARC stratum specifically; that identification is the open content of any follow-on phase (Section 6).

5.1. The B–T 1933 §§7–9 machinery. The Phase C verification [9, Phase C] extracts verbatim from the SHA-verified OCR text dump of [3] (PDF SHA-256 dcd7e3c6...d68fe6) the following three results:

- **§7 Theorem II** (factorization preservation): if $L_n = L_{n-r} L_r$ with L_{n-r}, L_r completely proper in a region, the product L_n is completely proper in the same region.
- **§9 Fundamental Theorem** (existence): every $L_n(y) = 0$ with coefficients of the kind specified in [3, §1] and known in the complete neighbourhood of infinity is completely proper in each of the several quadrants associated with the equation.
- **§11 Theorem III** (converse / sectorial realization): given a linearly independent set of formal series $e^{Q_j(x)} s_j(x)$ of the same general description as might occur in connection with a difference equation of order n , there exists a difference equation $L_n(y) = 0$ of [3, §1]-kind possessing the series as fundamental solutions.

5.2. Application to the SIARC stratum. The SIARC PCF Wallis recurrence (1) satisfies the [3, §1]-kind hypothesis: its coefficients a_n, b_n are polynomial in n (and in particular asymptotically rational in n at $n = \infty$), hence of the appropriate Gevrey-1 class. The order is $n = 2$ in the language of [3] (an order-2 linear difference operator); the polynomial degree d of the coefficient b_n enters only at the level of the leading symbol of the operator, not the [3, §1]-kind hypothesis itself.

The Phase C four sub-gates of [9, Phase C.2] (non-resonance match, non-degeneracy match, d -range applicability, μ -units consistency per the upstream A-01 verdict) all pass for the SIARC stratum at the *normal-case* locus $\deg a < 2 \deg b$ – with a *caveat* that the §7 Theorem II “point of division” condition $\Re Q_{\text{dom}}(x) > \Re Q_{\text{sub}}(x)$ *degenerates* on the borderline locus $\deg a = 2 \deg b$ (both formal exponents $\mu_{\text{dom}}, \mu_{\text{sub}}$ coalesce at $\deg b$, and the strict inequality on the real parts of the exponential dominants is replaced by an equality at leading order). The borderline-locus degeneracy is the literature-side counterpart of the borderline-locus mechanism (i') of Theorem 1.2.

5.3. Scope of the application. The application records that the formal-level analysis of Theorem 1.1 carries through the [3] §§7–9 existence and factorization machinery uniformly in $d \in [2, 8]$. It *does not* record that [3] identifies $A = 2d$ for the SIARC stratum specifically; the §9 Fundamental Theorem is universal in d but pin-points the formal-level μ -pair only via the [3, §1]-kind hypothesis, which gives the band $A_{\text{naive}} \in \{d-1, d, d+1\}$ already recorded in Theorem 1.1, not the lift to $A = 2d$. The lift is open in this note and is the open content of Section 6 Q6.1, Q6.2, and Q6.4.

6. OPEN QUESTIONS AND FOLLOW-ON PHASES

Open problem 6.1 (Mechanism identification). Which of the two mechanisms (i') borderline-locus and (ii') definitional of Theorem 1.2 is realised on the SIARC PCF Wallis stratum at $d \geq 3$? The [8] R1.1 + R1.3 + Q1 record is consistent with either; distinguishing them requires either (a) symbolic derivation of the $\sqrt{n}$ sub-leading correction predicted by mechanism (i') and an empirical test on the high-precision tail of $\log |p_n/p_{n-1}|$, or (b) a careful re-statement of the [8] fit-protocol to support or rule out the $A_{\text{fit}} = \mu_{\text{dom}} - \mu_{\text{sub}}$ identification under mechanism (ii').

Open problem 6.2 (Borderline-case algebraic ansatz). Under mechanism (i'), can the borderline-case algebraic ansatz $Q_j(x) = \pm B x^{1/2} + (\text{lower order})$ of [3, §1] be made explicit at the SIARC stratum, with B identified in closed form in terms of c_a and c_b ? The expected closed form is $B = \sqrt{c_a}$ at the formal level, by analogy with the $d = 2$ V_{quad} upper-branch transition of [7, §6]; an explicit derivation is open in this note.

Open problem 6.3 (Definitional mechanism). Under mechanism (ii'), what is the precise mathematical definition of A_{fit} extracted by the [8] high-precision tail-window fit? The fit is performed on $\log |p_n/p_{n-1}|$ at $\text{dps} = 80$; the linear-regression target is $n \log n$. Whether the extracted slope is $\mu_{\text{dom}} - \mu_{\text{sub}}$ or instead a Stokes-constant exponent (or a combination of the two) requires a fit-protocol audit.

Open problem 6.4 (Formal-to-analytic sectorial upgrade). The formal-level baseline of Theorem 1.1 is purely formal: it produces the two formal solutions $y_{\text{dom}}, y_{\text{sub}}$ as elements of $\Gamma(n)^\mu \gamma^n n^\rho \mathbb{C}[[1/n]]$, without an analytic-side guarantee that the corresponding formal Borel transforms converge in a sector of opening $> \pi/(2d)$ (the canonical sectorial-asymptotic existence condition for irregular singularities of formal rank $p = 1$). The formal-to-analytic sectorial upgrade is the content of the Wasow §X.3 (Theorem 11.1) [14] / Adams 1928 [1] / Turrittin 1955 [13] / Immink 1984 [5] framework, with the modern Borel-Laplace radius identification stated in [4, ch. 5] and the formal-side existence anchored in [2, §2] (see [9, Phase D] for the literature substrate). At the time of preparation of this note, the Wasow §X.3 OCR-readable PDF is unavailable on disk (image-only) and [1] has not been acquired (NIA per the upstream A-01 verdict of [9]). The sectorial upgrade is open and is the natural T1 Phase 3 follow-on.

AI DISCLOSURE

During the preparation of this work the author used GitHub Copilot (Microsoft) and Anthropic Claude to assist with code generation, numerical computations, and manuscript drafting. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of this article. AI assistants are not authors. All theorem statements (Theorem 1.1), proposition statements (Theorem 1.2), and conjecture statements (Theorem 1.3) remain the human author's responsibility, and every numerical or symbolic-derivation fact in this note traces to an AEAL entry in `claims.jsonl` with a SHA-256 output hash on the SIARC bridge under `sessions/2026-05-03/T1-BIRKHOFF-PHASE2-LIFT-LOWER/` (the Phase 2 substrate, commit 37c939f) and `sessions/2026-05-06/T1-PHASE2-BASELINE-NOTE/` (this version). The Birkhoff–Trjitzinsky 1933 §§7–9 verbatim extracts of Section 5 are anchored by the SHA-verified PDF dcd7e3c6...d68fe6 of [3] at the runbook-canonical literature directory (`tex/submitted/control center/literature/g3b_2026-05-03/`).

Related cross-cites for the Zenodo related-identifiers block. The present note's Zenodo deposit lists the following six SIARC Zenodo records as related identifiers (cite-block): PCF-1 v1.3 [7], PCF-2 v1.3 [8], Channel Theory v1.3 [10], T2B v3 [12], SIARC umbrella v2.0 [11], and D2-NOTE v2.1 [6].

APPENDIX A. BRIDGE SESSION PROVENANCE

The following SIARC bridge sessions are the AEAL anchors for the per-theorem claims in this note. Each is committed to the public `siarc-relay-bridge` GitHub repository at github.com/papanokechi/siarc-relay-bridge.

- `sessions/2026-05-03/T1-BIRKHOFF-PHASE2-LIFT-LOWER/` (commit 37c939f, 2026-05-04). Phase 2 of T1; verdict `UPGRADE_PARTIAL_FORMAL_LEVEL`. The substrate of this note. Phase A symbolic derivation `phase_a_symbolic_derivation.py` SHA-256 2fc6e392...f16248; Phase B extended sweep `phase_b_extended_sweep.py` SHA-256 39e98db6...8c10175; Phase C literature verification `phase_c_literature_verification.md` SHA-256 c6cc2b1a...808ae93; Phase D verdict `phase_d_verdict.md` SHA-256 7145a00d...2c65c5; Phase 2 `claims.jsonl` carries 16 AEAL entries.
- `sessions/2026-05-03/T1-A01-NORMALIZATION-RESOLUTION/`. Upstream A-01 normalization-resolution verdict, used in Section 5 to confirm μ -units consistency across Wasow / Birkhoff / Birkhoff–Trjitzinsky / Adams normalisation conventions; provides the SHA-verified [3] OCR text dump `.bt1933.txt` that supplies the verbatim extracts of the Phase C verification.

- `sessions/2026-05-06/T1-PHASE2-BASELINE-NOTE/` (this session). Drafting and Zenodo packaging of the present note. Contains `bt_baseline_note.tex`, `bt_baseline_note.pdf`, `annotated_bibliography.zenodo_description_bt_baseline_note.txt`, `zenodo_upload_bt_baseline_note_runbook.md`, `claims.jsonl` (≥ 10 AEAL entries), `rubber_duck_critique.md`, `verdict.md`, `halt_log.json`, `discrepancy_log.json`, `unexpected_finds.json`, and `handoff.md`.

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- [7] Papanokechi. PCF-1 v1.3: Stokes signal and channel scope for $\delta < 0$ degree-2 polynomial continued fractions, 2026. Version DOI 10.5281/zenodo.19937196; concept DOI 10.5281/zenodo.19931635. Section 6 Theorem 5 establishes the $A \in \{3, 4\}$ split at $d = 2$ in the α -direction Wallis stratum; the lower branch $A = 3$ is realised by the QL01–QL26 family of 26 representatives, with each representative’s A -value verified at $dps = 80$ across a high-precision tail-window fit. PDF SHA-256 63420dbf4abb7124672f522c37fc04ebdb3f6694ac39959456b2890d9788ff5e.
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- [9] Papanokechi. SIARC bridge session T1–BIRKHOFF–PHASE2–LIFT–LOWER: Phase 2 of the Birkhoff–Trjitzinsky lift for the SIARC PCF Wallis stratum, 2026. Bridge session at commit 37c939f, 2026-05-04. Verdict UPGRADE.PARTIAL.FORMAL.LEVEL: Phase A (`phase_a_symbolic_derivation.py`, SHA-256 2fc6e39267768791912cc53aa59bc231c40483a1a93e92104b13f07809f16248) + Phase B (`phase_b_extended_sweep.py`, SHA-256 39e98db607a5331686cf4ba4890bc50abc1232e8b533b80eeb2d4636c8e10175) establish the formal-level baseline $A_{\text{naive}} \in \{d - 1, d, d + 1\}$ across the three SIARC conventions for $d \in [2, 8]$; Phase C (`phase_c_literature_verification.md`, SHA-256 c6cc2b1ac437647f2cad974b07462cc43422eaa71cc079f8a76e3cc32808ae93) verifies the four C.2 sub-gates against Birkhoff–Trjitzinsky 1933 §§7–9. The Phase 2 `claims.jsonl` carries 16 AEAL entries; the present note carries forward 6 of them as literature-citation entries.
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